When Simple Is Enough: Leakage-Free Benchmarking of Classical Machine Learning vs. Graph Neural Networks in Low-Data Bioactivity Prediction⋆

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ABSTRACT

Small bioactivity data sets make it easy to overstate the benefit of complex models, through data leakage and through model selection that is rarely evaluated on unseen data. We re-examined this for inhibitors of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR, $N = 859$ curated compounds), comparing classical stacking ensembles on four molecular representations with a Directed Message Passing Neural Network (D-MPNN) on the same 25 folds of a repeated nested cross-validation. Ensembles on Morgan fingerprints reached $R^2 = 0.7592 \pm 0.0449$; adding 3D descriptors changed nothing detectable ($\Delta R^2 = 0.002$). The D-MPNN reached $R^2 = 0.7248$ with default settings and 0.7262 after a nested hyperparameter search, not significantly different from any classical representation after multiple-comparison correction (adjusted $p \geq 0.092$); equivalence with Morgan fingerprints held for the MAE but not for R^2. Feature selection and Gaussian-noise augmentation, credited with large gains in the original study, did not improve a classical learner, so those gains were not reproduced. With scaffold-grouped splits, R^2 fell by about 0.2 for both families. The classical benchmark was not cheaper to run than the network's. Applying the Morgan ensemble to an FDA-approved library (3378 compounds) and docking 17 candidates gave 4 consensus hits, 3 of which passed in silico ADMET filters; potency–docking agreement was weak ($\rho = -0.322$) and selectivity over human DHFR was not assessed. For this target, a fixed-fingerprint ensemble was sufficient; we make no claim beyond it.

Keywords: Quantitative Structure-Activity Relationship, Low-Data Regimes, Nested Cross-Validation, Data Leakage, Graph Neural Networks, Molecular Docking

1. Introduction

1.1. Predictive Modeling in Low-Data Bioactivity Regimes

Predictive modeling in quantitative structure-activity relationships (QSAR) operates under strict constraints when data is scarce Winkler (2026). Under these conditions, simpler interpretable approaches—such as multivariate linear regression or classical tree-based ensembles—frequently match or exceed the prediction reliability of complex artificial neural networks Shalit Peleg and Milo (2023); Xie et al. (2023). These parsimonious models maintain higher bias and lower variance. They resist the extreme sensitivity to random fluctuations that plagues overparameterized algorithms, particularly when supplied with robust, quantum-chemically derived descriptors Xie et al. (2023).

Small datasets inherently invite overfitting. This bottleneck worsens exponentially when the dimensionality of molecular descriptors exceeds the available sample size Hawkins (2004); Goodarzi et al. (2013). To circumvent data sparsity, contemporary workflows frequently deploy data augmentation (DAU) or synthetic sample generation to artificially expand the training manifold Gerz and Jelali (2026); Meyer et al. (2025). However, these interventions

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demand substantial computational overhead Meyer et al. (2025); Calle et al. (2025). More critically, improper DAU application obscures the true predictive capacity of the base algorithms. When augmentation precedes dataset partitioning (the "augment-then-split" fallacy), it violates independent and identically distributed (i.i.d.) assumptions. Models memorize synthetic artifacts and latent data dependencies rather than learning genuine physicochemical rules, yielding artificially inflated performance metrics Gerz and Jelali (2026); John et al. (2025); Veetil et al. (2024).

Evaluating complex architectures in these regimes requires strict separation between model building and model assessment. Hyperparameter tuning on small samples with large parameter spaces inevitably leads to model selection bias Cawley and Talbot (2010). Consequently, assessing machine learning architectures in isolation provides zero evidence of real-world utility unless the model selection procedure itself undergoes rigorous validation. Double (or nested) cross-validation is the accepted way to decouple model building from performance estimation Baumann and Baumann (2014); Cawley and Talbot (2010). Standard k-fold cross-validation systematically overestimates explorative capabilities Király et al. (2022); Xiong et al. (2020), and generative deep learning struggles to extrapolate within sparse, discrete chemical spaces like Simplified Molecular-Input Line-Entry System (SMILES) strings Liu et al. (2023). To preserve methodological integrity, any generative or augmentative process must remain strictly confined to the training folds after establishing an independent partition Gerz and Jelali (2026); John et al. (2025).

To safely harness overparameterized models Winkler (2026) without succumbing to the overfitting trap, empirical safeguards against chance correlation are non-negotiable. Techniques such as y-randomization (response shuffling) explicitly verify that the algorithm maps true chemical relationships rather than memorizing random noise Rücker et al. (2007). Although frequently recommended in literature, its empirical execution is routinely omitted in modern deep learning QSAR studies; running this check is therefore advisable before advancing candidates to virtual screening, as it makes chance correlation unlikely. When descriptors undergo meticulous curation and pass these randomization checks, machine learning effectively guides rational design—successfully discriminating potent bioactives, such as Amyloid-β aggregation inhibitors, from decoys Bansal et al. (2025).

Alternatively, transfer learning bypasses data scarcity by adapting base networks optimized on massive source domains Espley et al. (2023). By enforcing reduced learning rates during fine-tuning Bodun et al. (2025) and prioritizing efficient convolutional architectures over heavier Transformer models Özçelik and Grisoni (2024), researchers can navigate class imbalance and model selection bias in low-resource settings.

1.2. Domain-Engineered Features versus End-to-End Representation Learning

Traditional feature-based machine learning relies on domain-engineered descriptors and molecular fingerprints Das et al. (2026). Representations such as Morgan (ECFP4) encode structural topologies, while physicochemical and 3D descriptors capture specific molecular interactions deterministically without requiring massive training corpora Bansal et al. (2025); Iwuala et al. (2026); Xie et al. (2023). Embedding theoretical chemistry directly into the model ensures high interpretability and algorithmic transparency Xie et al. (2023). However, accumulating these handcrafted features generates highly dimensional, redundant matrices Goodarzi et al. (2013); Shalit Peleg and Milo (2023). In restricted datasets, this redundancy amplifies computational costs and exacerbates the feature selection bottleneck Goodarzi et al. (2013); Shalit Peleg and Milo (2023).

End-to-end representation learning bypasses manual feature selection. Deep learning paradigms—including Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Graph Neural Networks (GNNs)—autonomously map discrete molecular topologies into continuous latent spaces to model complex, nonlinear interactions Azad (2026); Das et al. (2026); Liu et al. (2023). Similarly, Chemical Language Processing (CLP) applies natural language techniques to string representations like SMILES Özçelik and Grisoni (2024). Yet, learning representations in discrete chemical spaces remains intrinsically fragile. Altering a single character in a SMILES string can irreparably invalidate the entire molecular structure Liu et al. (2023).

The utility of deep learning complexity in low-data regimes remains highly contested. These algorithms require dense data manifolds, rendering representation learning ineffective when ground-truth labels are scarce Konstantakos et al. (2025). To circumvent this limitation in *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) inhibition, recent studies deploy highly complex pipelines. Karimova and Senanayake (2025) employed Gaussian noise-based data augmentation and deep neural network (DNN) ensembles to stabilize training on a restricted dataset, elevating their baseline R^2 from 0.75 to 0.85. This intervention prompts a critical evaluation: is synthetic data generation strictly necessary to achieve predictive reliability? Parsimonious classical ensembles are reported to match or exceed the predictive fidelity of deep architectures in data-scarce scenarios, although such comparisons are rarely run on identical,

leakage-free folds Iwuala et al. (2026); Shalit Peleg and Milo (2023); Das et al. (2026); Dar et al. (2023); Espley et al. (2023); Xie et al. (2023).

Traditional skepticism toward nonlinear models in limited-data scenarios is evolving Dalmau et al. (2025). Properly regularized nonlinear algorithms, hybridized with structured linear priors, achieve competitive results comparable to robust classical methods Dalmau et al. (2025); Dar et al. (2023). Transfer learning serves as a primary catalyst for this shift. Fine-tuning pretrained networks like ChemBERTa allows researchers to extract accurate representation learning predictions using only dozens of target data points Oulladji et al. (2025); Espley et al. (2023).

1.3. Methodological Rigor and the Threat of Data Leakage in Molecular Modeling

Data leakage—the unintended introduction of test data information into the training process—is a statistically silent error that fundamentally undermines the validity of machine learning evaluations by inflating metrics without triggering code warnings Vrba et al. (2026); Gerz and Jelali (2026). In low-data QSAR modeling, this vulnerability frequently manifests across two primary vectors: preprocessing and model selection. When transformations like feature scaling, imputation, or variance-based filtering are applied globally across an entire dataset, the algorithm gains illicit knowledge of the test distribution. This violation of the independent and identically distributed (i.i.d.) assumption yields spuriously high performance estimates Correa et al. (2025); Lewis et al. (2023).

Similarly, traditional internal validation often suffers from model selection bias. If a validation set continuously influences hyperparameter tuning, it loses its statistical independence. Highly parameterized architectures, such as deep neural networks, are particularly prone to exploiting this overlap. They adapt to noise and produce deceptively optimistic metrics that severely underestimate true generalization errors Baumann and Baumann (2014); Konstantakos et al. (2025). To preserve methodological integrity, model selection and preprocessing must operate as strictly encapsulated components of the fitting process. Insufficient reporting of these partitioning mechanics often confounds scientific literature, leading researchers to incorrectly attribute high performance to novel complex architectures when the success actually stems from silent data leakage Rojas and Madanian (2026).

1.4. Objectives and Principal Contributions

This study asks whether, on one small and well-characterized bioactivity set (859 TgDHFR inhibitors), graph networks, 3D descriptors, feature selection and data augmentation improve on classical ensembles trained on fingerprints once the evaluation is free of leakage. We do not assume the answer; we design the evaluation so that either outcome would be detectable. Our contributions are:

- **Leakage-free, paired evaluation:** A reproducible Python pipeline built on a 5×5 repeated nested cross-validation protocol (25 outer evaluations) in which every preprocessing and selection step is fitted inside the training partitions, and in which the classical models and the D-MPNN are scored on exactly the same folds.
- **Calibrated statistical inference:** Use of the Nadeau–Bengio corrected t-test with Holm adjustment for the comparisons, which accounts for the overlap of training sets across repeated folds, and of an equivalence test (TOST) with a pre-specified margin, so that "no significant difference" is distinguished from "demonstrated equivalence".
- **Classical ensembles versus a graph network:** A comparison of four classical representations against a Chemprop D-MPNN, with default hyperparameters and with a hyperparameter search nested in the same folds, together with measured computational cost.
- **Replication of the original study's gains:** A test of whether the feature selection and Gaussian-noise augmentation reported by Karimova and Senanayake (2025) (from $R^2 = 0.75$ to 0.85 on a single hold-out split) transfer to a classical learner under nested cross-validation.
- **Robustness checks:** Repetition of the benchmark under scaffold-grouped splits and with the extreme-potency records retained, to quantify how much the conclusions depend on the splitting scheme and on the curation.
- **Screening as hypothesis generation:** Application of the final model to FDA-approved drugs, followed by molecular docking and in silico ADMET filters, to produce a short list of candidates. These are computational hypotheses; none was tested experimentally, and selectivity against human DHFR was not assessed.

2. Materials and Methods

An overview of the multi-tier virtual screening pipeline implemented in this study is illustrated in Figure 1.

2.1. Bioactivity Benchmark Dataset and Curation

This study utilizes a previously published benchmark of *Toxoplasma gondii* dihydrofolate reductase (TgDHFR) enzyme inhibitors Karimova and Senanayake (2025). Adopting the baseline dataset compiled by Karimova and Senanayake establishes a rigorous methodological control. This framework isolates the predictive capacity of classical machine learning algorithms from computationally heavy statistical interventions. Omitting synthetic data augmentation during the primary benchmarking phase ensures that performance metrics reflect empirical ground-truth data rather than algorithmic artifacts.

Initial records merged drug-protein interaction parameters and SMILES structures from BindingDB with bioactivity data from ChEMBL Karimova and Senanayake (2025). Automated standardization protocols, including salt stripping and structural validation, ensured chemical integrity Bodun et al. (2025). The merged table contained 875 curated records. Extreme potency values were screened with Tukey's rule on their pIC_{50} distribution (fences at Q1 $-$ 1.5 IQR $=$ 2.06 and Q3 $+$ 1.5 IQR $=$ 10.80). 2 records fell outside the fences: a classical antifolate analogue at $pIC_{50} = 11.68$ (high fence) and a practically inactive compound at $pIC_{50} = 1.56$ (low fence). They were excluded from the primary analysis, which leaves 873 entries, the size reported in the original study Karimova and Senanayake (2025). Because an exclusion of this kind can itself shape the reported performance, the full benchmark was also run with these two records retained (Section 3.7).

For compounds exhibiting multiple IC_{50} measurements across varying experimental conditions, we applied the Interquartile Range (IQR) method. Duplicate assays with a bioactivity variance exceeding 1.5 log units were discarded as statistical outliers Karimova and Senanayake (2025). The remaining consistent entries were averaged to yield a single representative measurement per ligand.

To adhere to the Organization for Economic Co-operation and Development (OECD) principles for QSAR validation Tropsha (2010); Cherkasov et al. (2013); Iwuala et al. (2025) and to keep the same structure out of both training and test folds, all SMILES were then canonicalized with RDKit. The 873 entries turned out to describe only 859 distinct structures: 14 molecules appeared twice under different PubChem CIDs, and the second occurrence of each was removed (the first was retained). The measured pIC_{50} values within each pair differed by at most 0.18 units, so the target distribution is practically unchanged. The removed records are listed in `data/V2-combined_bd_chmbl_unique_duplicates.csv`.

This curation protocol yielded a final benchmark of 859 unique small molecules. The continuous target variable was transformed into pIC_{50} ($-\log_{10}(IC_{50})$) Karimova and Senanayake (2025). This logarithmic conversion standardizes response values and compresses the active range, optimizing the target distribution for robust regression modeling.

To explicitly quantify the performance impact of the interventions reported in the original literature, we replicated its hold-out protocol in an isolated ablation study. A 15% hold-out partition was separated first (seed 42), and everything below was learned from the development partition only. The baseline (A) is the stacking ensemble on all 2048 Morgan bits. The feature-selection variant (B) computes the permutation importance of every bit on the development partition (5 repeats, negative mean squared error) and keeps the bits whose mean importance is positive beyond floating-point noise (greater than 10^{-9}), because bits that the model never uses have an importance that is zero up to rounding error. The augmentation variant (C) adds, to the selected features, perturbation of the training features via additive Gaussian noise: applying two distinct noise levels ($\sigma = 0.01$ and $\sigma = 0.001$) generated two synthetic variants per original molecule, which effectively triplicated the training space (to 2190 rows) while maintaining absolute isolation of the empirical test set. Each variant is the average of five stacking ensembles that differ in the seed of their tree models.

2.2. Representation Spaces: Molecular Fingerprints and Conformational Descriptors

We generated four distinct molecular representation spaces using RDKit Landrum et al. (2026) to systematically map the structure-activity relationship. The structural baseline relied on Morgan fingerprints (ECFP4). We computed these with a radius of 2 and folded them into a fixed-length vector of 2048 bits. Extended-connectivity fingerprints offer rapid calculation and capture a vast array of distinct local substructural features Rogers and Hahn (2010); chirality was not encoded. Classical 2D fingerprint-based models frequently achieve predictive parity with state-of-the-art 3D structure-based models when relying exclusively on ligand information Gao et al. (2020).

To establish a physicochemical baseline, the pipeline extracted the complete RDKit descriptor list (217 conventional two-dimensional topological and physicochemical descriptors). Curated 2D molecular representations

demonstrate substantial robustness; recent studies confirm they maintain scaffold-independent advantages over 3D alternatives across various property prediction tasks Tiwari et al. (2026). A third hybrid space concatenated these 217 descriptors with the 2048-bit Morgan fingerprints to evaluate potential synergistic effects.

A fourth representation appended ten 3D shape descriptors (asphericity, eccentricity, inertial shape factor, the two normalized principal moment ratios, the three principal moments of inertia, radius of gyration, and spherocity index) to this hybrid, for 2275 features in total, to assess the predictive value of spatial geometry. Hydrogens were added and a single conformer per molecule was generated with the Experimental-Torsion Knowledge Distance Geometry algorithm (ETKDGv3, fixed random seed of 42, random initial coordinates as a fallback) Riniker and Landrum (2015). Because purely distance-based constraints often distort aromatic rings and sp^2 centers, ETKDGv3 integrates experimental torsion-angle preferences derived from small-molecule crystallographic data. The conformer was then locally minimized with the MMFF94 force field before descriptor extraction. Conformer embedding failed for 0 of the 859 molecules, and 0 descriptor values of the 2D+3D+FP matrix were missing or infinite before imputation.

Although integrating spatial information remains critical for molecular image-based modeling Matsuzaka and Uesawa (2023) and geometry-augmented representation learning Zhang and Bai (2024), recent empirical evaluations suggest that 3D conformer descriptors yield limited incremental predictive value over curated 2D representations in standard tabular benchmarks Tiwari et al. (2026). Our nested cross-validation framework explicitly tested this hypothesis. The purely structural Morgan fingerprints yielded a mean R^2 of 0.7592, without a significant difference from the 3D hybrid space ($R^2 = 0.7680$; Nadeau-Bengio-corrected paired test on the per-fold MAE, $t = 2.030$, $p = 0.0536$). The two best-scoring spaces, the 2D descriptors with fingerprints and the 3D hybrid, were likewise indistinguishable from each other ($p = 0.7505$). Since no space was reliably superior and the Morgan space needs neither descriptor calculation nor conformer generation, it was adopted for all downstream feature selection, data augmentation, and virtual screening protocols.

2.3. Nested Cross-Validation Protocol and Variance Correction

Evaluating machine learning architectures in low-data regimes demands rigorous isolation between model optimization and performance estimation. Standard cross-validation partitions datasets into k disjoint blocks, systematically rotating them between training and testing phases Calle et al. (2025); Krstajic et al. (2014). While conventional nested cross-validation (NCV) literature frequently defaults to symmetrical 5×5 or 10×10 architectures Calle et al. (2025); Lewis et al. (2023); Wainer and Cawley (2021), methodological frameworks do not mandate identical fold counts for inner and outer loops Krstajic et al. (2014). To guarantee leak-free evaluation, balance computational overhead, and mitigate the model selection bias described in Section 1.3, we implemented a strict 5×5 repeated nested cross-validation protocol for this restricted dataset ($N = 859$). This architecture executes a 5-fold outer loop repeated 5 times (random seed 42), embedding a 5-fold inner loop, mathematically decoupling model assessment from model selection. The same outer partitions were used for every molecular representation, which allows paired comparisons between them.

Data partitioning strategies dictate the validity of small-scale QSAR models Király et al. (2022). Bemis-Murcko scaffold splitting is widely advocated to test out-of-domain extrapolation and prevent structural leakage Yang et al. (2019); Basit Iqbal et al. (2025); Iwuala et al. (2026). However, given the restricted size of this benchmark ($N = 859$), scaffold-based partitioning is expected to generate imbalanced folds, in which entire bioactivity ranges could vanish from the training pool. Our primary objective is interpolation within the established applicability domain of known TgDHFR inhibitors, rather than extrapolation to entirely novel chemotypes, so random splitting was adopted for the main benchmark, which keeps the bioactivity distribution consistent across folds. Because random splits are known to be optimistic for series of close analogs, we also repeated the Morgan-fingerprint ensemble and the D-MPNN under scaffold-grouped splits. The Bemis-Murcko scaffold of each compound (without stereochemistry) defined a group; the 859 compounds fall into 262 groups, the largest with 49 members, and 159 groups contain a single compound. In each of the 5 repeats, groups were visited in random order and each was assigned to the currently smallest of the 5 folds, so that no scaffold appears in both the training and the test partition of an outer fold. Everything else (the 25 outer folds, the inner loop with random folds, the hyperparameters and the corrected tests) was identical to the random-split protocol, and both model families again shared the same folds.

The outer loop of the NCV framework evaluates generalization. To achieve lower variance in performance estimation compared to extreme approaches like Leave-One-Out (LOO) Krstajic et al. (2014), we evaluated 25 outer splits. Standard parametric tests assume independent samples. Repeated cross-validation violates this assumption via overlapping training sets across iterations. To neutralize this intra-repetition correlation, we applied the Nadeau and

Bengio variance correction Nadeau and Bengio (1999) across all downstream statistical comparisons: the variance of the mean difference over the $J = 25$ outer folds is inflated from $\hat{\sigma}^2/J$ to $(1/J + n_{test}/n_{train})\,\hat{\sigma}^2$, with $n_{test}/n_{train} = 1/4$ for the 5-fold outer loop. In each iteration, one fold is strictly held out as the test set while the remaining pool passes into the inner loop. This absolute isolation ensures the test data remains hidden during feature and model selection, yielding a nearly unbiased out-of-sample prediction estimate Varma and Simon (2006); Wainer and Cawley (2021).

The inner loop governed architecture selection. Within each outer training pool, an embedded 5-fold inner loop evaluated all 41 single, pairwise, and three-way stacking combinations (up to 3 base estimators) and retained the combination with the highest mean inner-fold R^2. These combinations were drawn from six diverse regressor families: Random Forest (RF), Extra Trees (ET), Light Gradient Boosting Machine (LightGBM), eXtreme Gradient Boosting (XGBoost), Support Vector Regression (SVR with RBF kernel, $C = 10$), and k-Nearest Neighbors (kNN, $k = 5$, cosine distance). The tree ensembles used 200 trees; all hyperparameters were fixed in advance and none was tuned, so that the inner loop selects among architectures only. For multi-model combinations, a `StackingRegressor` utilizing `RidgeCV` as the meta-estimator, trained on 5-fold out-of-fold base predictions, coordinated the meta-learning phase. The selected combination was then refit on the whole outer training pool and scored once on the held-out outer fold. Predictions were clipped to the training range extended by 3 pIC_{50} units as a safeguard; this affected 0 predictions across all outer folds.

This inner loop also serves as a structural safeguard against preprocessing leakage. Executing global transformations—such as z-score standardization, median imputation, or supervised feature selection—prior to fold partitioning inadvertently leaks dataset-wide statistics into the training phase Cawley and Talbot (2010); Varma and Simon (2006). To eliminate this vulnerability, every transformation was implemented as a step of the model pipeline, so that its parameters were computed only on the training partition at hand (an inner training fold during selection, the outer training pool for the final fit) and subsequently applied to the validation or test data without re-estimation Vrba et al. (2026). For the continuous descriptor spaces, the pipeline chained median imputation, standardization (`StandardScaler`), and feature selection that kept every descriptor with a strictly positive importance in a Random Forest (`SelectFromModel`) before the regressor; the binary Morgan space was passed to the regressors unchanged. By confining model selection and these transformations entirely within the training partitions, we prevent inadvertent information leakage. This strict encapsulation guarantees that reported metrics reflect genuine generalizability rather than selection artifacts Parvandeh et al. (2020); Lewis et al. (2023); Calle et al. (2025).

2.4. Deep learning baseline: Directed message passing neural networks

Benchmarking classical machine learning ensembles against advanced representation learning required implementing a Directed Message Passing Neural Network (D-MPNN) via the Chemprop framework (version 2.2.4) Graff et al. (2025); Yang et al. (2019). To guarantee methodological parity, the D-MPNN was evaluated on the same 25 outer partitions (5×5 repeated cross-validation, random seed 42) used for the classical baselines, with each molecule predicted by a model that never saw it. Two variants were run: a default configuration, for which no inner loop was needed, and a tuned variant with a hyperparameter search nested inside each outer fold (see below). This alignment keeps the performance metrics directly comparable and free from single-split variance artifacts.

The architecture processes molecular SMILES strings by mapping them into graph representations, designating atoms as vertices and bonds as edges Heid et al. (2023). Initial feature vectors capturing atomic identity and topological connectivity were constructed using `SimpleMoleculeMolGraphFeaturizer()`. The network applied `BondMessagePassing()` (hidden size 300, depth 3, no dropout) to extract embeddings directly from directed edges rather than vertices. This edge-centric design eliminates spurious feedback loops, reducing noise in the learned molecular representations Yang et al. (2019). A `MeanAggregation()` function pooled these atomic embeddings into a unified molecular vector.

Within each training partition, the continuous pIC_{50} targets were standardized with the training mean and standard deviation, and predictions were transformed back to the original scale. The pooled vector was batch-normalized and fed into a `RegressionFFN()` (one hidden layer of 300 units) to yield the final predictive output. The mean squared error on the standardized targets was the training loss. Model weights were optimized via PyTorch Lightning for a fixed number of 100 epochs on CPU, using Chemprop's default learning-rate schedule (2 warm-up epochs, initial rate 10^{-4}, peak 10^{-3}, final 10^{-4}) and a per-fold random seed (42 plus the fold index). No early stopping or checkpoint selection was applied, so the model after the last epoch was used and the held-out fold never influenced training.

Unlike the classical ensembles, whose architecture was selected by an inner search, the default D-MPNN used Chemprop's default hyperparameters (hidden size, depth and dropout as above). Quantitatively, one D-MPNN training

took on average 204 seconds (range 197–211 s) on the workstation CPU, so the 25 folds required about 85 minutes. A search matching the classical one, with 205 trainings per outer fold (41 architecture combinations across 5 inner splits), would take roughly 12 days for the full protocol at that training time, which is why the tuned variant uses a reduced search.

The tuned variant searched a grid of 8 configurations (hidden size 300 or 600, message-passing depth 3 or 5, dropout 0 or 0.2; the default configuration is one of them). Inside each outer training pool, 20% of the compounds were held out at random, every configuration was trained on the remaining 80% and the one with the lowest root-mean-square error on the held-out part was retained. That configuration was then retrained on the whole outer training pool, with the same 100-epoch schedule and per-fold seed, and scored once on the outer test fold. The held-out outer fold therefore played no role in the choice. The search is deliberately smaller than the classical one (a single inner split instead of 5 folds, and 8 configurations instead of 41 combinations), so it tests whether a moderate amount of tuning helps rather than the best result that exhaustive tuning could reach. One tuned outer fold took on average 2192 s, and the complete tuned benchmark amounted to 15.2 h of worker time (two training processes ran in parallel). For comparison, a classical outer fold including its complete inner search took between 398 and 833 seconds on average, depending on the representation (wall-clock times on the same machine, not normalized by thread count). The default variant isolates the intrinsic inductive bias of the graph architecture from the confounding artifacts of small-sample tuning Cawley and Talbot (2010); Shalit Peleg and Milo (2023), and the tuned variant checks whether that choice handicapped the network.

Out-of-sample performance was quantified using the coefficient of determination (R^2) and Mean Absolute Error (MAE) averaged across all 25 folds. This protocol yielded a final cross-validated test performance of $R^2 = 0.7248 \pm 0.0439$ and an MAE of 0.4748, establishing the deep learning baseline for subsequent comparative analysis.

2.5. Virtual Screening Protocol and Ligand Efficiency Profiling

An automated virtual screening pipeline, driven by the feature-selected Morgan-fingerprint stacking ensemble of the augmentation study (Random Forest, XGBoost, and SVR combined by a ridge meta-learner, trained without augmentation on the 730-compound development partition of the 15% hold-out split; hold-out $R^2 = 0.7347$), systematically prioritized viable lead candidates from an FDA-approved library ($N = 3378$). Rather than relying solely on raw predicted bioactivity, the evaluation framework integrated applicability domain enforcement with multi-objective ligand efficiency profiling. This approach mitigates the risk of prioritizing "molecularly obese" compounds that exhibit high theoretical potency but possess unfavorable physicochemical properties for downstream development Hopkins et al. (2004).

Initial structural pre-filtering restricted the FDA library to compounds containing only H, C, N, O, F, P, S, Cl, Br, and I, a set that covers all but 1 of the training compounds. This constraint prevents the model from extrapolating into unsupported chemical spaces. A size ceiling (molecular weight $\leq$ 1000 Da; the largest training compound weighs 879 Da) removed very large molecules whose potency the model could not meaningfully extrapolate.

To enforce the applicability domain quantitatively, an `IsolationForest` algorithm (5% contamination threshold, fixed seed) was fitted on the fingerprints of all 859 curated compounds, restricted to the same selected Morgan bits (1255 of 2048) used by the deployed ensemble, and evaluated the remaining FDA compounds. This unsupervised anomaly detection excluded compounds lying in sparsely populated regions of the training fingerprint space, so that all advanced candidates resided within the region of chemical space covered by the known bioactivity data. Ultimately, this hierarchical funnel reduced the initial library of 3378 compounds to 2516 after the size and elemental filters, to 1967 inside the applicability domain, and finally to 17 candidates for orthogonal docking validation: the 10 top-ranked compounds by predicted potency (which already included methotrexate, pyrimethamine, and triamterene from the candidate list of the original study) plus 7 further reference entries from that list (trimethoprim as base, sulfate, hydrochloride, and sulfamethoxazole combination, finerenone, etodolac, and bisacodyl). As a consistency check, the ensemble recovered the experimental pIC_{50} of the reference inhibitors pyrimethamine (predicted 6.54 versus 6.56) and trimethoprim (5.71 versus 5.57); both compounds belong to the model's training partition, so this verifies fit and not generalization.

Four distinct efficiency metrics were then computed to contextualize predicted activity for physicochemical load. Ligand Efficiency (LE) quantified the binding energy per heavy atom count (HAC) Hopkins et al. (2014):

$$\mathrm{LE} = \frac{1.37 \times pIC_{50}}{\mathrm{HAC}} \tag{1}$$

To account for heteroatom contributions and total mass, the Binding Efficiency Index (BEI) substituted HAC with molecular weight (MW) Abad-Zapatero and Metz (2005):

$$\mathrm{BEI} = \frac{pIC_{50}}{\mathrm{MW}} \times 1000 \tag{2}$$

Furthermore, to balance potency against lipophilicity and mitigate the risk of nonspecific binding, Lipophilic Ligand Efficiency (LLE) was computed as the difference between predicted potency and the calculated partition coefficient (LLE = pIC_{50} − cLogP, Crippen method in RDKit). Finally, the Surface Efficiency Index (SEI) evaluated potency relative to the topological polar surface area (TPSA), providing a proxy for membrane permeability potential (SEI = (pIC_{50}/TPSA) × 100). Because composite indices combining logarithmic potency values with linear physical parameters can introduce size-dependent mathematical artifacts Shultz (2013), these four metrics (LE, BEI, LLE, SEI) were computed from the predicted pIC_{50} and reported alongside it for interpretation; candidates were ranked and selected by predicted potency, and none of the metrics was used as a filter.

2.6. Orthogonal Physics-Based Validation via Molecular Docking

Orthogonal structure-based evaluation via AutoDock Vina v1.2.7 Das et al. (2026); Trott and Olson (2010) complemented the ligand-based predictions. This physical validation translates abstract chemical representations into spatial coordinates, mitigating the risk of false positives induced by algorithmic scaffold memorization. By assessing structural complementarity with the binding site, the docking protocol acts as an orthogonal plausibility filter. Vina employs an Iterated Local Search optimizer and a hybrid scoring function, calibrated against PDBbind data, to guide flexible ligands into optimal binding conformations Eberhardt et al. (2021).

Three-dimensional receptor coordinates were retrieved from the Protein Data Bank (*Toxoplasma gondii* DHFR, PDB ID: 6AOG) wwPDB consortium (2019); Berman et al. (2003). The receptor was prepared as a rigid apo structure: all non-protein records (crystallographic waters, ions, cofactors including the bound NADPH, and the co-crystallized ligand) were removed, retaining both protein chains. The structure was converted to the PDBQT format with OpenBabel in rigid mode Babel (2011), keeping heavy atoms only: no hydrogens were added and no partial charges were assigned, which the Vina scoring function does not use. While excluding the NADPH cofactor omits a potential source of active-site polarization, this apo-receptor approximation is a common simplification in structure-based virtual screening. The Vina search space was centered on the coordinates of the native CP6 ligand in chain B ($x = 3.689$, $y = 39.992$, $z = -62.818$) with a cubic grid box of $20.0 \times 20.0 \times 20.0$ Å and an exhaustiveness parameter of 32, which permitted unrestricted exploration of ligand orientations and rotatable bonds within the site Trott and Olson (2010).

Prior to screening the machine learning candidates, the docking protocol was checked by re-docking the co-crystallized CP6 ligand into the TgDHFR active site. The top-ranked pose reproduced the crystallographic position with a centroid distance of 0.32 Å, below the 2.0 Å criterion adopted for validation. This criterion compares the geometric centers of the docked and crystallographic ligands and is therefore more lenient than a heavy-atom RMSD, which also tests orientation; moreover, the search box was centered on the crystallographic ligand. The check thus confirms that the grid parameters and the docking setup are consistent with the known binding site, but it does not demonstrate blind pose prediction.

Following validation, Vina evaluated all 17 candidates of the screening step (the 10 top-ranked compounds plus the reference entries). For each compound the largest fragment was retained to remove counter-ions, hydrogens were added, a single 3D conformer was generated with ETKDGv3 (fixed seed) and minimized with MMFF, and a flexible-ligand PDBQT file was prepared with Meeko, keeping the protonation state of the input structure. Docking used a fixed random seed and returned the 3 best poses. Poses were ranked by the predicted binding free energy (ΔG) in kcal/mol, where lower energies denote stronger predicted binding Feng et al. (2023); this is an empirical, rigid-receptor estimate and not a measured thermodynamic quantity. The docking ligand efficiency was computed as the best score per heavy atom. To systematically evaluate the consensus between the ligand-based and structure-based approaches, we established quantitative thresholds. Compounds were classified as validated consensus hits only if they exhibited both high ML-predicted potency ($pIC_{50} > 6.5$) and strong docking affinity ($\Delta G < -8.0$ kcal/mol). Compounds meeting the ML threshold but failing the docking criterion were flagged as potential ML false positives. Conversely, compounds with $pIC_{50} < 6.0$ were labeled validated negatives if $\Delta G > -7.5$ kcal/mol and potential ML false negatives if $\Delta G < -8.0$ kcal/mol; all remaining compounds were classed as neutral. Finally, to quantify the monotonic agreement between the two orthogonal methods without assuming linearity, a Spearman rank correlation analysis was conducted between the continuous pIC_{50} predictions and the Vina docking scores (Section 3.4).

2.7. In Silico Pharmacokinetic (ADMET) Profiling

Following the docking step, a predictive pharmacokinetic (ADMET) assessment was applied to all 17 docked candidates, not only to the consensus hits, to flag late-stage attrition risks. All structures were reduced to their largest fragment before the calculations, so that counter-ions of salt forms (e.g., glucuronate or hydrochloride) did not distort the descriptor counts or the fingerprints. Candidates first underwent evaluation against three criteria of Lipinski's Rule of Five, eliminating molecules with more than 5 hydrogen bond donors, more than 10 hydrogen bond acceptors, or a calculated LogP (Crippen method) above 5; the molecular-weight criterion was not applied because the screening had already capped the molecular weight. To assess critical toxicity and absorption endpoints, models were trained on datasets from the Therapeutics Data Commons (TDC): 655 compounds for hERG and 910 for Caco-2 permeability (Caco2-Wang). Molecular structures were encoded as 2048-bit Morgan fingerprints (radius 2) and processed through Random Forest models comprising 100 estimators with a fixed random seed.

A `RandomForestClassifier` trained on the TDC hERG dataset evaluated cardiac toxicity risk. This served as a safety filter, advancing only compounds with a predicted probability of hERG channel blockade below 0.5. Unlike this exclusion criterion, a `RandomForestRegressor` trained on the Caco2-Wang dataset used intestinal permeability predictions to categorize the likely administration route. Compounds with moderate-to-high predicted permeability ($\log P_{app} > -5.15$, P_{app} in cm/s) were classified as orally compliant. Conversely, candidates falling below this threshold were not discarded; they were retained and flagged as parenteral candidates, for which poor intestinal absorption is not a limitation.

These models are screening-level estimates rather than experimental toxicity or absorption data, and their predictive performance was not assessed here. Some candidates are also present in the training sets (of the 15 compounds evaluated, 6 appear in the hERG set and 1 in the Caco-2 set), so their predictions are not independent. Hepatic metabolism, off-target toxicity and blood-brain barrier penetration, which matters for a parasite that infects the brain, were not modeled. The profile therefore prioritizes compounds for further evaluation and does not by itself establish safety.

2.8. Computational Environment and Reproducibility

To support algorithmic transparency and computational reproducibility Rojas and Madanian (2026), all data preparation, model training, and evaluation steps were programmatically implemented in Python (version 3.11.16) and chained in a single entry point that runs the 20 steps of the study, from the construction of the training sets to the final consistency check of the manuscript. The computational environment relied on a standardized core software stack:

- **Cheminformatics and Featurization:** RDKit (version 2025.9.3) Landrum et al. (2026) executed all molecular parsing, structural standardization, and descriptor generation (2D, 3D, and ECFP4); Open Babel (version 3.1.1) Babel (2011) converted the receptor to PDBQT.
- **Classical Machine Learning:** The nested cross-validation framework Krstajic et al. (2014); Calle et al. (2025) and baseline algorithms leveraged `scikit-learn` (version 1.6.0) Pedregosa et al. (2011), `xgboost` (version 3.1.2) Chen and Guestrin (2016), and `lightgbm` (version 4.6.0) Ke et al. (2017). Data manipulation, matrix operations, and statistical tests were managed via `pandas` (version 2.3.3) McKinney et al. (2010), `numpy` (version 1.26.4) Harris et al. (2020), and `scipy` (version 1.16.3).
- **Deep Learning Infrastructure:** The D-MPNN graph models utilized `chemprop` (version 2.2.4) Graff et al. (2025); Heid et al. (2023) and `lightning` (version 2.6.5), with underlying tensor operations supported by PyTorch (`torch`, version 2.5.1) Paszke et al. (2019).
- **Docking and ADMET:** AutoDock Vina (Python bindings, version 1.2.7) Trott and Olson (2010) performed the docking, ligands were prepared with Meeko (version 0.8.0), and the ADMET datasets were loaded through PyTDC (version 0.4.1).
- **Hardware Specifications:** All model training, architecture selection, and evaluation protocols ran on an Ubuntu 26.04 LTS workstation with two Intel(R) Xeon(R) E5-2695 v2 processors (24 physical cores, 48 threads, 2.4 GHz) and 247 GB of RAM, using CPU only; no GPU was involved.

All random number generators were seeded (seed 42). The models of the augmentation study were fitted single-threaded, because a multi-threaded random forest sums its trees in a thread-dependent order and changes the last bits of its predictions; parallelism was applied across independent tasks (features in the permutation importance, runs in the ensemble) so that the results do not depend on the number of workers. Together with the numerical tolerance of the selection rule described above, this made the feature selection reproducible: permutation importances computed with 46 and with 8 workers were bitwise identical. The graph-model benchmark is not bit-reproducible, since multi-threaded CPU training of the network varies slightly between executions (mean R^2 between 0.725 and 0.729 across three executions). A complete execution from scratch required approximately 30 hours on the workstation, dominated by two nested cross-validation runs of about 13 hours each (main and sensitivity), the y-randomization test (about 2.7 hours), and the graph-model benchmark (about 85 minutes).

3. Results

3.1. Comparative Performance of Classical ML Across Feature Spaces

We established a baseline for TgDHFR inhibitor prediction by evaluating classical machine learning ensembles across four molecular representations with a 25-fold nested cross-validation framework (Figure 2). RDKit 2D descriptors combined with fingerprints reached $R^2 = 0.7700 \pm 0.0467$ (MAE = 0.4305), the 3D hybrid space $R^2 = 0.7680 \pm 0.0437$ (MAE = 0.4315), and the purely structural Morgan fingerprints (ECFP4, 2048 bits) $R^2 = 0.7592 \pm 0.0449$ (MAE = 0.4477). RDKit 2D descriptors alone were clearly inferior ($R^2 = 0.7225 \pm 0.0492$, MAE = 0.4711).

Differences between representations were assessed with Nadeau-Bengio corrected paired t-tests on the shared outer folds. The three fingerprint-containing spaces could not be separated in R^2 (Morgan versus 2D descriptors with fingerprints, $p = 0.2297$; Morgan versus the 3D hybrid, $p = 0.3118$; 2D descriptors with fingerprints versus the 3D hybrid, $p = 0.7505$). In MAE, Morgan trailed the two hybrid spaces by 0.017 and 0.016 pIC_{50} units ($p = 0.0475$ and $p = 0.0535$), a small gap that is no longer significant after Holm adjustment for the six comparisons ($p_{\mathrm{Holm}} = 0.1425$ for both). In contrast, 2D descriptors alone were significantly worse than every space that includes fingerprints (versus 2D descriptors with fingerprints, $p = 0.0003$; versus the 3D hybrid, $p = 0.0004$; versus Morgan, $p = 0.0059$). Topological fingerprints are therefore the primary driver of predictive capacity, and the ten 3D shape descriptors derived from ETKDGv3 embedding and MMFF94 minimization added no measurable advantage for this benchmark.

To rule out chance correlation, a 100-iteration y-randomization test was applied to the ET+XGB+SVM stacking ensemble (the most frequently selected architecture of the best-scoring representation) trained on Morgan fingerprints and scored by 5-fold cross-validation. With the true potencies this model reached $R^2 = 0.7638$ (a single 5-fold partition, which explains the small difference from the nested estimate), whereas the permuted models collapsed to a mean R^2 of -0.0135 ± 0.0084, a separation of $Z = 92.6$ standard deviations. The ensemble therefore maps genuine structure–activity relationships rather than memorizing noise.

Selection of the stacking architecture was unstable (Figure 3). Within the Morgan space, Support Vector Regression entered the selected ensemble in 92% of the outer folds and Random Forest in 60%, followed by XGBoost (32%), LightGBM (24%), k-nearest neighbors (24%), and Extra Trees (12%); the selected ensembles contained one, two, or three models in 12%, 32%, and 56% of the folds. No single hierarchy dominated: the most frequent exact combination, RF+SVM, won only 32.0% of the folds (8/25), followed by RF+XGB+SVM (3/25) and 9 further architectures selected in at most 2 folds each. The descriptor-based spaces behaved differently: Extra Trees was selected in 100%, 100%, and 100% of the folds of the 2D, 2D+FP, and 2D+3D+FP spaces, respectively, and SVR in 64%, 44%, and 40%. While aggregated performance remains stable across outer folds, the data sparsity of the 859-compound benchmark prevents the inner loop from isolating a universally superior ensemble.

We then placed these results against the incremental progression reported in the original study (Table 1). Our classical stacking ensemble trained solely on Morgan features ($R^2 = 0.7592$) performed comparably to their unselected 2D/3D/FP baseline ($R^2 = 0.75$); under a Nadeau-Bengio corrected one-sample t-test the difference was not significant ($t = 0.382$, $p = 0.7062$). In contrast, the literature value for the model with permutation-importance feature selection ($R^2 = 0.82$) lies significantly above our nested cross-validated Morgan estimate ($t = -2.514$, $p = 0.0191$), and so does the final data-augmented deep neural network ensemble ($R^2 = 0.85$; $t = -3.755$, $p = 0.0010$). This comparison has a limitation: the original values come from a single hold-out split without a reported dispersion and are treated here as fixed constants, so the test does not capture the sampling variability of that split. Achieving a competitive baseline

Table 1
Methodological comparison isolating the performance impact of architectural complexity, descriptor dimensionality, and data augmentation in TgDHFR bioactivity prediction.

Model Framework	Descriptors	Data Augmentation	Test R^2 (mean ± st
Current Study (Classical Ensemble)	ECFP4 (2D)	None	0.7592± 0.0449
Baseline Karimova and Senanayake (2025)	2D/3D/FP	None	0.75[a]
Feature-selected model Karimova and Senanayake (2025)	2D/3D/FP (Selected)	None	0.82[a]
Augmented Final Model Karimova and Senanayake (2025)	2D/3D/FP (Selected)	Gaussian Noise	0.85[a]

[a] Standard deviation not reported (NR) in the original study.

($R^2 \approx 0.75$) does not require computationally expensive 3D descriptors or complex neural network architectures; simple 2D fingerprints paired with classical algorithms suffice.

To test whether the interventions credited with the gains of the original study transfer to classical learners, we replicated its protocol (15% hold-out, seed 42; 129 molecules held out before any selection or augmentation, 730 for development) on a stacking ensemble (Figure 4). The unselected baseline reached $R^2 = 0.7341$ (MAE = 0.4378). Permutation-importance selection (1255 of 2048 bits retained) yielded $R^2 = 0.7347$ (MAE = 0.4386), and Gaussian-noise augmentation yielded $R^2 = 0.7132$ (MAE = 0.4724). Paired bootstrap differences showed no significant effect of either intervention: selection changed R^2 by +0.0006 (95% CI [−0.0021, +0.0034]) and augmentation by −0.0209 (95% CI [−0.0534, +0.0161]) relative to the baseline, and both intervals include zero. The hold-out intervals are wide, spanning roughly 0.2 R^2 units, and contain the original baseline of 0.75; the original values of 0.82 and 0.85 lie above the upper limits of the intervals for selection ($p_{\text{boot}} = 0.0450$) and augmentation ($p_{\text{boot}} = 0.0010$), respectively. The gains reported for the original deep ensemble were therefore not reproduced by feature selection or augmentation applied to a classical learner; they may depend on the neural architecture, or reflect the sampling variability of a single hold-out split.

3.2. Benchmarking Tree-Based Ensembles Against Graph Neural Networks

Graph neural networks learn molecular representations directly from the atom–bond graph, but their advantage is expected to shrink when labeled data are scarce. To test whether representation learning helps on the TgDHFR dataset (859 compounds), a Directed Message Passing Neural Network (D-MPNN, Chemprop) was evaluated on exactly the same 25 outer folds as the classical ensembles, in two variants described in the Methods: default hyperparameters, and a hyperparameter search nested inside each outer fold.

The default D-MPNN reached a mean fold R^2 of 0.7248 ± 0.0439 and a mean absolute error of 0.4748, compared with $R^2 = 0.7592 \pm 0.0449$ for the Morgan-fingerprint stacking ensemble (Figure 5). Because the folds are shared, the models can be compared with the same Nadeau–Bengio corrected paired t-test used for the classical representations. The network was numerically below the Morgan ensemble ($\Delta R^2 = -0.0344$) and the fingerprint-based hybrid ensembles ($\Delta R^2 = -0.0452$ and -0.0433 against 2D+FP and 2D+3D+FP), and on par with the 2D descriptors alone ($\Delta R^2 = +0.0023$), but no difference was significant after Holm adjustment (smallest adjusted p: 0.127 for R^2 and 0.062 for MAE).

Tuning did not change this picture. In the nested search, the selected configuration varied across folds (7 of the 8 configurations were chosen at least once, and the default one in only 4% of the folds), but the tuned network reached $R^2 = 0.7262 \pm 0.0342$ and MAE = 0.4703, indistinguishable from the default one ($\Delta R^2 = +0.0014$, $p = 0.924$; $\Delta\text{MAE} = -0.0044$, $p = 0.704$). Against the classical ensembles the tuned network was again numerically lower than Morgan ($\Delta R^2 = -0.0330$, Holm-adjusted $p = 0.124$) and the two fingerprint-based hybrids ($\Delta R^2 = -0.0438$ and -0.0419), and on par with the 2D descriptors ($\Delta R^2 = +0.0037$); none of these differences was significant (smallest adjusted p: 0.092 for R^2 and 0.080 for MAE).

Equivalence is a separate question. With the ±0.05 margin, the tuned network was equivalent to the 2D descriptors in R^2 (90% CI [−0.032, +0.039], $p = 0.018$), but equivalence to the Morgan ensemble could not be shown for R^2 (90% CI [−0.062, −0.004], $p = 0.163$). In MAE, the tuned network and the Morgan ensemble were equivalent within ±0.05 ($\Delta = +0.0226$, 90% CI [−0.003, +0.048], $p = 0.041$). The evidence therefore supports that the network is not clearly worse in absolute error, but leaves open a deficit in R^2 as large as $\Delta R^2 = -0.062$. The tuned search was deliberately modest (Methods), so a larger search or a different architecture could change this.

Table 2
Top 10 FDA-approved candidates predicted by the classical ML model, ranked by predicted pIC_{50} and evaluated across standard ligand efficiency metrics.

Rank	Compound Name	Predicted pIC_{50}	LE	BEI	LLE	SEI
1	Methotrexate	7.4879	0.31	16.48	7.22	3.56
2	Trimetrexate Glucuronate (CID 54949)[a]	7.3813	0.25	13.10	7.93	2.92
3	Trimetrexate Glucuronate (CID 336805)[a]	7.3813	0.25	13.10	7.93	2.92
4	Pralatrexate	6.7783	0.27	14.20	5.80	3.27
5	Triamterene	6.7531	0.49	26.66	5.92	5.21
6	Pyrimethamine	6.5352	0.53	26.28	4.01	8.40
7	Folic Acid	6.2649	0.27	14.19	6.31	2.94
8	Bendamustine Hydrochloride	6.0557	0.35	15.34	2.37	10.38
9	Erdafitinib	6.0254	0.25	13.49	1.84	7.79
10	Brigatinib	6.0092	0.21	10.29	0.92	7.00

[a] Duplicate PubChem CID entries mapping to the same underlying structure in the FDA library.

Both graph-network variants and the classical ensembles fall in the same range as the original study's unselected baseline ($R^2 = 0.75$). The final augmented ensemble of the original study ($R^2 = 0.85$) lies above this range, but, as discussed in Section 3.1, that value comes from a single hold-out split and was not reproduced by selection or augmentation on a classical learner. Under the data volume of this benchmark, the graph architecture therefore did not provide a measurable gain over fixed fingerprints.

Computational cost does not favor the classical approach in this protocol, but the comparison needs care. One default D-MPNN training took about 204 s on the workstation CPU (about 85 minutes for the 25 folds), whereas a classical outer fold with its complete inner search took between 398 and 833 s depending on the representation (614 s, about 4.3 h in total, for Morgan). The default network received no model selection. When it did, in the tuned variant, one outer fold took 2192 s, about 15.2 h of worker time for the whole benchmark, which is more than the classical protocol even though the network's search was smaller. The case for the classical ensembles therefore rests on accuracy that is not lower, on simpler inputs and models that run on a CPU, and on a deployed model that uses only a subset of the fingerprint bits, and not on a shorter benchmark.

3.3. Virtual Screening and Prioritization of FDA-Approved Candidates

The deployed model, the Morgan-fingerprint stacking ensemble described in Section 2.3 (nested-CV $R^2 = 0.7592$), was applied to the FDA-approved library after the funnel described in the Methods: 3378 compounds were reduced to 2516 by the size and elemental filters and to 1967 by the Isolation Forest applicability domain. Two known inhibitors served as a consistency check. Pyrimethamine was predicted at pIC_{50} 6.54 (experimental 6.56, absolute error 0.02) and trimethoprim at 5.71 (experimental 5.57, error 0.14). Both compounds belong to the training partition, so this verifies that the final model fits its training data and says nothing about generalization; the generalization estimate remains the nested cross-validation reported above.

Table 2 lists the ten compounds with the highest predicted potency, and Figure 6 places them against ligand efficiency (LE). The antifolates methotrexate, trimetrexate (listed twice, under two PubChem entries that map to the same structure) and pralatrexate occupy the top of the ranking by absolute potency. Methotrexate is also part of the training set, so its rank partly reflects memorization; trimetrexate and pralatrexate are not. Absolute potency, however, scales with molecular size, and these large molecules have modest efficiency (LE between 0.25 and 0.31, with a LE of 0.30 commonly taken as the lower bound of a good lead).

Size-corrected metrics reorder the list. Pyrimethamine, which is also in the training set, and triamterene, which is not, combine predicted pIC_{50} values of 6.75 and 6.54 with the highest LE (0.49 and 0.53) and BEI (26.7 and 26.3) of the table, well above the antifolates. Triamterene is therefore the clearest repurposing candidate from the ML stage: a predicted potency close to that of pyrimethamine at a comparable atomic efficiency, and which is absent from the training data. The remaining top-ranked compounds have predicted pIC_{50} between 6.0 and 6.3, and all but bendamustine have an LE below 0.30.

Table 3
Predicted ADMET profile of the 4 consensus hits and of the known inhibitors pyrimethamine and trimethoprim. Structures that appear under several entries are listed once. hERG prob. is the predicted probability of channel blockade (flag above 0.5) and Caco-2 $\log P_{app}$ the predicted permeability (oral if above -5.15).

Compound	Pred. pIC_{50}	ΔG (kcal/mol)	hERG prob.	Caco-2 $\log P_{app}$	Profile
Pralatrexate	6.78	−9.50	0.31	−5.41	Parenteral
Methotrexate	7.49	−8.71	0.40	−5.87	Parenteral
Trimetrexate Glucuronate	7.38	−8.48	0.40	−4.86	Oral
Triamterene	6.75	−8.04	0.80	−4.66	hERG flag
Pyrimethamine	6.54	−7.36	0.12	−4.42	Oral
Trimethoprim	5.71	−7.27	0.14	−4.77	Oral

3.4. Integrated ML-docking consensus filtering and false-positive mitigation

Neither ligand-based prediction nor docking is reliable alone, so the 17 candidates of the screening step were scored by both and classified by agreement (Figure 7). A compound is a consensus hit when it is predicted active by the ML model ($pIC_{50} > 6.5$) and docks favorably ($\Delta G < -8.0$ kcal/mol); discordant cases are reported, not resolved.

The global agreement between the two methods is weak. Spearman's rank correlation between predicted pIC_{50} and Vina score was $\rho = -0.322$ ($p = 0.2072$), with the expected sign (higher predicted potency, lower and therefore better docking energy) but no statistical significance. The estimate is also less informative than its $N = 17$ suggests: the four trimethoprim entries share one structure and one docking score, and the two trimetrexate entries are the same compound under two PubChem CIDs, so the effective number of independent compounds is smaller. We therefore do not interpret the correlation as evidence of orthogonality or of agreement; it only shows that docking does not simply reproduce the ML ranking.

Of the 17 candidates, 5 entries (4 unique compounds) were consensus hits, 4 were concordant negatives (all trimethoprim entries), 1 was an ML false positive, 1 an ML false negative, and 6 fell between the thresholds. The consensus hits are pralatrexate ($pIC_{50} = 6.7783$, $\Delta G = -9.50$ kcal/mol), methotrexate (7.4879, -8.71), trimetrexate glucuronate (7.3813, -8.48, both CIDs) and triamterene (6.7531, -8.04). Pralatrexate has the best docking score of the whole set. Methotrexate belongs to the training set, so its consensus status is the least informative; pralatrexate, trimetrexate and triamterene have no exact structural match in the training set.

The discordant cases show the limits of each method. Pyrimethamine, an established TgDHFR inhibitor, was predicted at $pIC_{50} = 6.5352$ (it is in the training set, so this is a fitted value) but docked at only $\Delta G = -7.36$ kcal/mol; trimethoprim, a weaker known inhibitor, scored similarly ($\Delta G = -7.27$ kcal/mol). Small ligands such as these obtain modest Vina scores because the score grows with the number of contacts, which suggests a limited sensitivity of the scoring function for small inhibitors rather than an ML artifact. Bisacodyl is the opposite case: docking favored it ($\Delta G = -8.76$ kcal/mol) while the model predicted $pIC_{50} = 4.8617$. Bisacodyl has none of the diaminopyrimidine-like features of the antifolate ligands in the training set, so a docking artifact of a flexible, hydrophobic ligand is the more plausible explanation, but without experimental data we cannot arbitrate between the two methods.

These results are a computational prioritization. The receptor is rigid, no water or protonation states are modelled, and Vina scores are empirical estimates, so the consensus hits are hypotheses for experimental testing, not confirmed inhibitors.

3.5. In Silico ADMET Profiling of the Docked Candidates

All 17 docked entries were profiled (Section 3.4; the entries include duplicates of the same structure, such as the four trimethoprim forms and the two trimetrexate CIDs). Of these, 15 satisfied the three Lipinski criteria. Among these, the hERG classifier flagged 6 entries as likely channel blockers (predicted probability above 0.5), leaving 9 entries: 7 classified as oral candidates and 2 as parenteral candidates because of low predicted Caco-2 permeability (Figure 8). Table 3 shows the profile of the consensus hits of Section 3.4 together with the two known inhibitors.

Of the 4 consensus hits, 3 passed both filters. Pralatrexate and methotrexate had a low hERG probability (0.31 and 0.40) but low predicted permeability ($\log P_{app}$ of −5.41 and −5.87), so they were classified as parenteral. Trimetrexate was predicted to be hERG-safe and orally absorbed. Triamterene, the most ligand-efficient candidate of the ML screen, was the only consensus hit removed: its predicted hERG probability was 0.80. The known inhibitor pyrimethamine (hERG probability 0.12) and trimethoprim were both classified as oral and hERG-safe, consistent with their use as oral

Table 4
Cross-validated performance (mean ± SD over 25 outer folds) under random and scaffold-grouped splits. The drop is the difference in mean R^2.

Model	R^2 (random)	R^2 (scaffold)	Drop in R^2	MAE (scaffold)
Morgan ensemble	0.7592 ± 0.0449	0.5778 ± 0.1015	0.1814	0.6096
D-MPNN	0.7248 ± 0.0439	0.5260 ± 0.1212	0.1988	0.6428

Table 5
Nested cross-validation performance without (primary analysis, $n = 859$) and with ($n = 861$) the two extreme-potency records. p-values compare the two runs with a Welch test using the Nadeau-Bengio variance correction Nadeau and Bengio (1999); the runs use different fold partitions, so folds are not paired.

	R^2				MAE			
Representation	Without	With	Δ	p	Without	With	Δ	p
RDKit 2D FP	0.770	0.744	-0.026	0.46	0.431	0.451	+0.021	0.43
RDKit 2D+3D FP	0.768	0.747	-0.021	0.54	0.432	0.449	+0.018	0.48
Morgan FP	0.759	0.746	-0.013	0.70	0.448	0.458	+0.010	0.68
RDKit 2D	0.723	0.704	-0.018	0.64	0.471	0.487	+0.016	0.57

drugs. These predictions come from Random Forest models trained on public data, whose accuracy was not evaluated here, and some of the compounds are in their training sets (Section 2.7), so the profile is a coarse prioritization and not a safety assessment.

3.6. Robustness to Scaffold-Grouped Splits

Random folds place close analogs of a test compound in the training set, which favors models that memorize local structure. To measure how much of the reported accuracy depends on this, the Morgan-fingerprint ensemble and the D-MPNN were re-evaluated with scaffold-grouped outer folds (Section 2.3), keeping the 25 folds, the hyperparameters and the shared partitions. Table 4 summarizes the outcome.

Both families lost about 0.2 units of R^2 when scaffolds were held out: from 0.7592 to 0.5778 for the Morgan ensemble (MAE from 0.4477 to 0.6096) and from 0.7248 to 0.5260 for the D-MPNN (MAE from 0.4748 to 0.6428). The fold-to-fold spread also more than doubled (SD of R^2 from 0.0449 to 0.1015 and from 0.0439 to 0.1212, respectively), because the difficulty of a fold now depends on which scaffolds it contains. The random-split estimates of the main benchmark therefore overstate the performance expected on new chemotypes.

The comparison between families kept its direction but not its resolution. Under scaffold splits the D-MPNN was lower than the Morgan ensemble by $\Delta R^2 = -0.0518$ (corrected $p = 0.205$, Holm-adjusted; $\Delta\text{MAE} = +0.0332$, $p = 0.236$). The 90% confidence interval of the R^2 difference, $[-0.120, +0.016]$, includes zero but also extends beyond the ± 0.05 equivalence margin, and the equivalence test was not significant ($p = 0.518$). With the present data we can thus neither claim a difference nor equivalence between the two families on new scaffolds; the sample (859 compounds) is too small to separate them given the larger variance of this protocol.

3.7. Sensitivity to Extreme Potency Values

The two records excluded by the Tukey rule (Section 2.1) were retained in a second, complete execution of the nested cross-validation benchmark ($n = 861$ versus $n = 859$ in the primary analysis; identical algorithms, search space and 5×5 protocol). If these records were informative members of the chemical series, retaining them should not degrade generalization; if they are outliers, retaining them should lower accuracy. Table 5 reports the outcome.

Retaining the two records lowered R^2 in all four representations (from -0.026 to -0.013) and raised the MAE (from $+0.010$ to $+0.021$ pIC_{50} units), a consistent direction that is nevertheless indistinguishable from fold-to-fold variability (p between 0.43 and 0.70). For the Morgan representation, R^2 moved from 0.759 to 0.746 ($p = 0.70$). The ordering of the three best representations changed between runs, but their differences were already within noise in the primary analysis, so the central conclusion (no representation is reliably superior) holds in both.

To test whether the records are predictable at all, a Morgan fingerprint Random Forest trained on the primary set was asked to predict them. The record at $pIC_{50} = 11.68$ was predicted as 6.59 (absolute error 5.09) and the record at 1.56 as 4.98 (error 3.42), that is, 8.2 and 5.5 times the out-of-fold RMSE of that model (0.62), and larger than every out-of-fold error among the 859 retained compounds.

Taken together, the two records behave as outliers with respect to the model: they cannot be anticipated from the structure–activity relationship learned from the rest of the series, and keeping them modestly worsens, without significantly altering, the reported metrics. Their exclusion is therefore justified as a pre-specified applicability-domain decision rather than as a way of improving performance, and it does not affect any comparison reported in this study. We note that the high-potency record may be a genuine measurement of a very potent antifolate rather than an error; in either case it lies outside the domain that the present data can support.

4. Discussion

4.1. The Role of Explicit Inductive Bias in Small-Scale Molecular Datasets

Graph neural networks can learn molecular features directly from graph topology, which removes the need for hand-designed descriptors, but this flexibility is usually paid for with training data. In our benchmark, with only 859 compounds, the D-MPNN reached $R^2 = 0.7248$ with default hyperparameters and 0.7262 after a nested hyperparameter search, numerically below the Morgan-fingerprint ensemble ($\Delta R^2 = -0.0344$ and -0.0330, respectively) and statistically indistinguishable from every classical representation after multiple-comparison correction (Section 3.2). Tuning itself changed nothing detectable ($\Delta R^2 = +0.0014$), and the configuration chosen by the search varied from fold to fold. In this dataset, therefore, learning the representation brought no detectable gain over using a fixed one.

A plausible reading is that explicit chemical priors matter when labels are scarce. A circular fingerprint hands the learner a fixed, chemically motivated description of substructures, whereas a message-passing network must learn an equivalent description from a few hundred molecules and a noisy target. We stress that this is an interpretation, not a result of this study: no experiment isolates the inductive bias. The network search was modest (8 configurations, one inner split), and, although it did not help, a larger search, a pretrained network or a different architecture could close the gap; our data cannot say otherwise. Equivalence with the Morgan ensemble was shown for the absolute error but not for R^2 (Section 3.2), so a small deficit of the network cannot be ruled out.

The augmentation experiments argue against one common remedy. The original study attributed part of its gains to Gaussian-noise augmentation and permutation-importance feature selection applied to a deep ensemble. Applied to a classical stacking ensemble under the same hold-out protocol, neither intervention improved on the unaugmented baseline (Figure 4): selection changed R^2 by $+0.0006$ and augmentation by -0.0209, with confidence intervals that include zero (Section 3.1). This shows that the gains of the original study did not transfer to a classical learner. It does not show that augmentation is useless for neural networks, which we did not test, and it does not by itself explain the original gains, which may also reflect the variability of a single hold-out split.

The practical conclusion is limited to this regime. For a dataset of this size, a classical ensemble on Morgan fingerprints is a sound default: it performed at least as well as the graph network we tested, tuned or not, with simpler inputs and fewer moving parts. Whether the balance changes with more data is an open question that these results cannot address; larger datasets are generally expected to favor learned representations, but we did not measure a learning curve, and the point at which a graph network would overtake fixed fingerprints is likely to depend on the target and on how well the network is tuned.

4.2. Dimensionality, Conformational Overhead, and Model Efficiency

Richer molecular representations are only worth their cost if they add predictive information. Three-dimensional descriptors are the most expensive to obtain, since they require conformer embedding (ETKDGv3) and force-field minimization (MMFF) for every molecule. Our comparison separates what each block of features contributes (Section 3.1). The 2D physicochemical descriptors alone were clearly the weakest representation ($R^2 = 0.7225$, against 0.7592 for Morgan fingerprints; $\Delta R^2 = 0.037$, Holm-adjusted $p = 0.0234$). Adding fingerprints to them recovered the difference ($R^2 = 0.7700$), and adding the ten 3D descriptors on top changed nothing detectable ($R^2 = 0.7680$; $\Delta R^2 = 0.002$, $p = 0.7505$). Morgan fingerprints alone were within 0.009 of the 2D+3D+FP set in R^2 (Holm-adjusted $p = 0.6891$). The hybrid sets had a slightly lower MAE than Morgan alone ($\Delta = 0.016$), but this was not significant after correction ($p = 0.1425$). In this dataset the fingerprint block carries the predictive signal, and neither the descriptors

nor the 3D block added a detectable amount to it. This conclusion concerns accuracy on this target; a ten-descriptor 3D summary could miss information that a richer 3D representation would capture.

The cost side is less clear-cut than usually assumed. For the 859 compounds, computing the features took 1.3 s for Morgan fingerprints and 5.3 s for the full 2D+3D+FP set on matrices of 2048 and 2275 features, respectively (wall-clock time with parallel workers on the workstation). The 3D set is the more expensive, but the absolute difference is a few seconds. The cost of a complete outer fold, with its inner architecture search, was instead dominated by model fitting: between 398 and 833 s on average across the four representations (614 s for Morgan), without a clear relation to whether 3D descriptors were present. At this scale, then, conformer generation was not the bottleneck. We did not measure how the cost of 3D features grows for libraries of millions of compounds, where it could matter, and the deployed screening model does not use them: it takes only the 1255 selected Morgan bits, which are computed from the 2D structure.

The parsimonious choice for this target is therefore the 2D fingerprint ensemble. This is a statement about information content and about the inputs the screening step needs, not a measured speed advantage; as shown in Section 3.2, the end-to-end benchmark of the classical protocol was not faster than the graph network.

4.3. Methodological Scope, Limitations, and Translational Potential

The reliability of any quantitative structure–activity relationship (QSAR) model is bounded by the chemical diversity of its training data. With 859 compounds, the applicability domain of the ensemble is restricted to the scaffolds and physicochemical ranges of the curated TgDHFR set, and the Isolation Forest filter used in the screening only excludes compounds that are clearly outside it. The cross-validation estimate also needs a caveat: the main folds were random, and scaffold-grouped splits (Section 3.6) lowered R^2 by about 0.2 for both the Morgan ensemble (from 0.7592 to 0.5778) and the D-MPNN (from 0.7248 to 0.5260). The reported R^2 therefore describes interpolation among related compounds, and the expected accuracy on the structurally distant drugs of the FDA library is closer to the scaffold-split values. We did not evaluate temporal splits. The predicted potencies of those drugs have no experimental counterpart in this study.

Activity cliffs are a specific weakness of 2D circular fingerprints in this setting: very similar molecules can differ strongly in activity, and a Morgan bit vector may map them to nearly the same representation. We did not measure how many cliffs the data set contains or how the model behaves on them. The docking step does not remove the problem either; it adds an independent estimate, but a rigid-receptor Vina score is itself unreliable for small differences between close analogs, as the discordance for pyrimethamine in Section 3.4 illustrates.

The inner architecture search is a further source of instability. Evaluating 41 stacking combinations per outer fold carries a risk of selection bias. The nested design keeps this search away from the outer test folds, so the estimate is not inflated by it, but the chosen architecture varied across folds: within the Morgan space, RF+SVM was the most frequent choice yet was selected in only 32.0% of the 25 folds (Figure S1). With a data set of this size the inner loop does not converge on a single architecture, and the reported performance should be read as that of the selection procedure and not of one fixed ensemble.

The ADMET step (Section 3.5) shows how much the final list depends on the filters: triamterene, the most efficient candidate of the ML screen, was removed on a predicted hERG probability of 0.80 from a Random Forest trained on 655 compounds whose accuracy we did not evaluate, so the flag is a reason to test the compound and not a reason to discard it. The parenteral label given to pralatrexate and methotrexate comes from a passive-permeability model and should not be read as a statement about clinical use: methotrexate, for example, is also given orally, because its absorption is carrier-mediated and is not captured by a passive-permeability predictor.

The most important translational limitation is selectivity. Pralatrexate, methotrexate and trimetrexate are anticancer antifolates that inhibit human dihydrofolate reductase, and are cytotoxic to host cells. Our pipeline predicts potency against the parasite enzyme and ranks docking energies against it, but it does not assess selectivity over the human enzyme, so a high rank should not be mistaken for a therapeutic candidate. The consensus hits are best read as computational hypotheses: compounds whose predicted activity is supported by two independent methods and that merit selectivity docking against human DHFR and experimental testing. The same pipeline could be applied to other targets with small data sets, but our results do not show that it transfers; that would need to be tested on those targets.

4.4. Conclusions and Future Directions

This study compared classical machine learning ensembles with a graph neural network on a small TgDHFR data set ($N = 859$), under a leakage-free nested cross-validation protocol, and carried the best classical model through a docking and ADMET triage of FDA-approved drugs. Three conclusions follow.

First, on this data set a classical stacking ensemble on 2D Morgan fingerprints performed at least as well as the alternatives we tested. It reached $R^2 = 0.7592 \pm 0.0449$, against 0.7248 ± 0.0439 for a default D-MPNN and 0.7262 ± 0.0342 for one tuned in a nested search; the difference ($\Delta R^2 = -0.0330$ for the tuned network) was not significant after multiple-comparison correction, and equivalence held for the MAE but not for R^2. Adding 3D descriptors to the fingerprint-based sets changed nothing detectable. Feature selection and Gaussian-noise augmentation, which the original study credited with large gains, did not improve a classical learner under the same hold-out protocol, so those gains were not reproduced. These findings are limited to a single target and a comparison in which the network search was modest; they do not show that graph networks cannot match fixed fingerprints. The classical protocol was also not cheaper to run: the benchmark took longer than the network's, and its advantage lies in accuracy that is not lower, simpler inputs, and a deployed model that needs only a subset of fingerprint bits.

Second, nested cross-validation and a careful curation of the data (removal of duplicates and of two extreme potency values, with a sensitivity analysis of the latter) were necessary to obtain estimates that can be compared across models. Our random-split estimates are also optimistic: with scaffold-grouped splits R^2 fell by about 0.2 for both model families (to 0.5778 for the Morgan ensemble and 0.5260 for the D-MPNN), and the data could not resolve the difference between them.

Third, combining ML prediction, docking and ADMET produced a short, reproducible list of hypotheses and not validated leads. Of the 17 compounds advanced to docking, 4 unique compounds were consensus hits, and 3 of them passed the ADMET filters: pralatrexate, methotrexate and trimetrexate. The agreement between predicted potency and docking energy was weak ($\rho = -0.322$, $p = 0.2072$), and the most ligand-efficient candidate, triamterene, was flagged by the hERG model. All surviving compounds are antifolates that also inhibit the human enzyme, and selectivity was not assessed.

Future work should therefore (i) dock the consensus hits against human DHFR and test them in enzymatic assays against TgDHFR, (ii) evaluate temporal splits, (iii) compare against larger searches and pretrained graph models, and (iv) apply the pipeline to other targets before claiming that it generalizes.

Data and Code Availability

All source code, Python scripts for dataset processing, nested cross-validation, model training, and docking validation protocols, as well as the benchmark datasets, are publicly available in the GitHub repository at the following link

Declaration of Generative AI's use

During the preparation of this work, the author(s) utilized generative artificial intelligence technologies (Large Language Models) solely to assist with language polishing, structural formatting, and improving the overall readability of the manuscript. After using these tools, the author(s) thoroughly reviewed, edited, and validated all generated text. The author(s) take full and sole responsibility for the final content, scientific integrity, and accuracy of this publication. No AI technologies were used to generate data, perform statistical analyses, or draw scientific conclusions.

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CRediT authorship contribution statement

Andrey Vinajera-Zamora: Conceptualization, Methodology, Formal analysis, Software, Validation, Investigation, Writing – original draft, Visualization. **Juan Castillo-Garit:** Validation, Writing – review & editing . **Samuel Montejo-Sánchez:** Validation, Writing – review & editing, Supervision. **Raul Arias-Carrasco:** Validation, Supervision, Writing – review & editing , Visualization.

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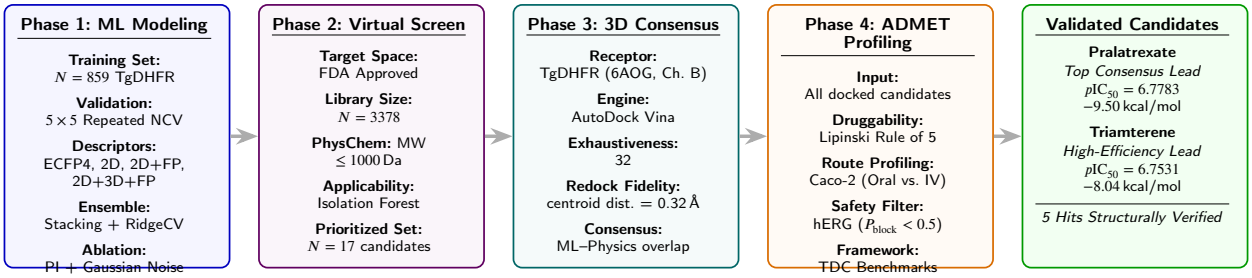


Figure 1: Schematic overview of the multi-tier computational repurposing and validation pipeline. The hierarchical funnel integrates: (1) 2D ligand-based representation benchmarking under nested cross-validation, (2) applicability domain-constrained screening of FDA-approved compounds, (3) orthogonal physics-based consensus validation via molecular docking (AutoDock Vina), and (4) in silico ADMET profiling (Caco-2 permeability route categorization and hERG cardiotoxicity filtering) to isolate clinically viable TgDHFR candidates.

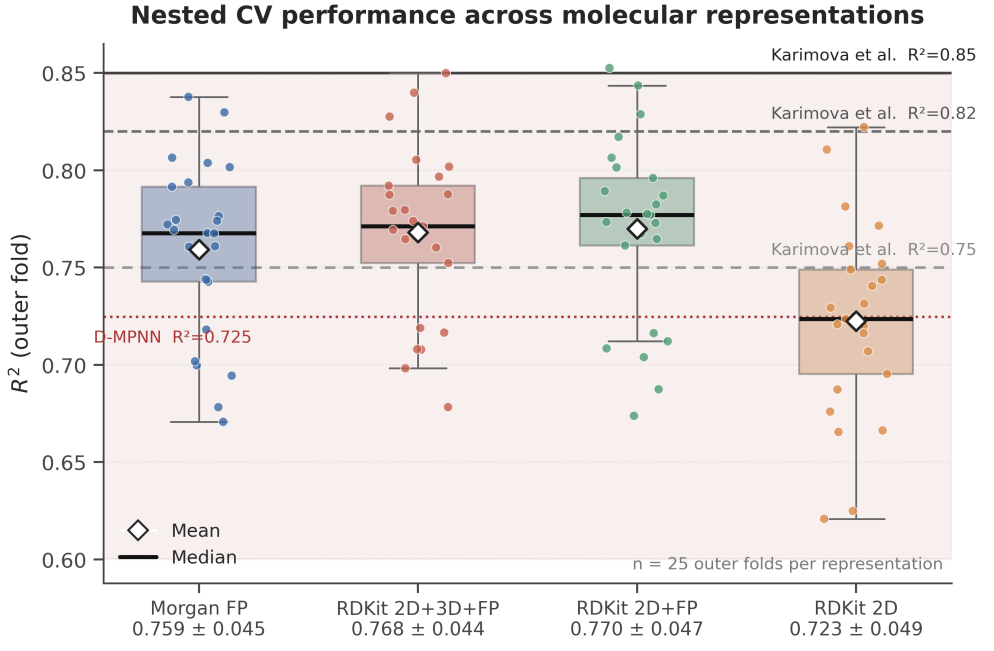


Figure 2: Nested cross-validation R^2 across four molecular representations (each point is one of the 25 outer folds). Horizontal lines mark the three values reported in the original study Karimova and Senanayake (2025); the dotted line and band show the mean ± SD of the D-MPNN benchmarked in Section 3.2. The three representations that contain fingerprints are statistically indistinguishable in R^2.

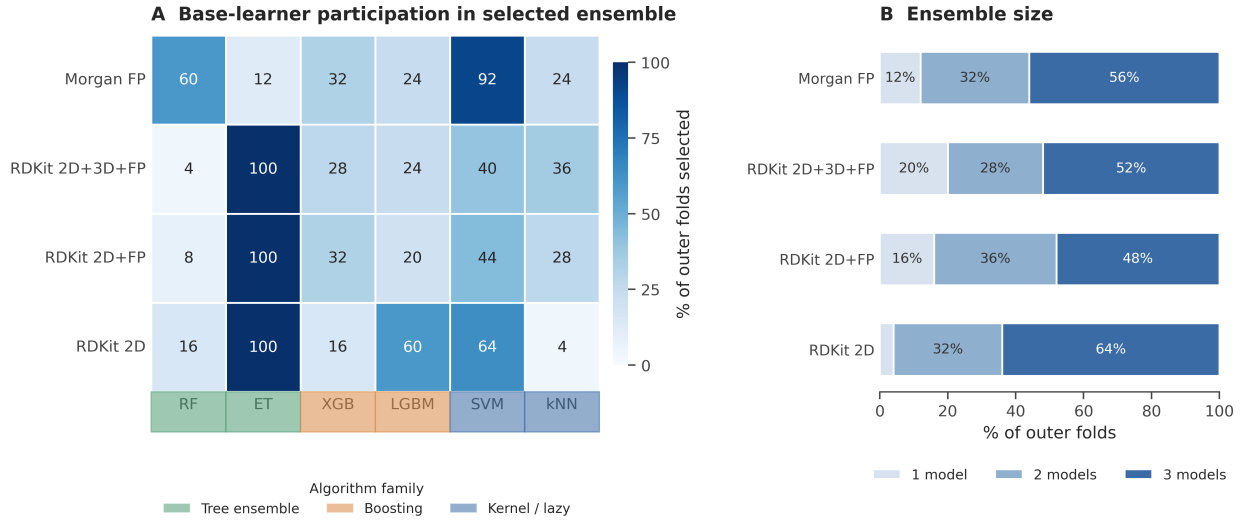


Figure 3: Composition of the stacking ensembles selected by the inner loop in the 25 outer folds. (A) Percentage of folds in which each base learner entered the selected ensemble, per representation. (B) Distribution of ensemble size (one, two, or three base models). No single architecture dominates in the Morgan space.

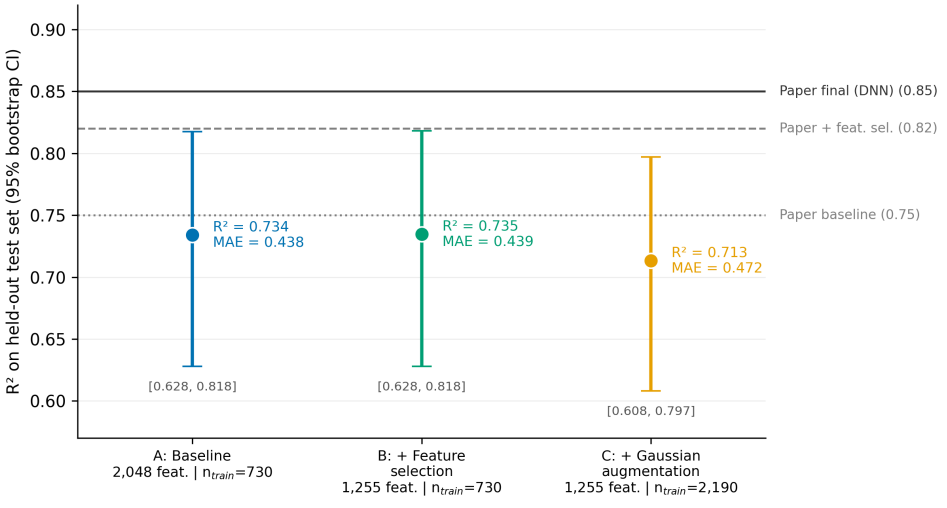


Figure 4: Replication of the original hold-out protocol on a classical stacking ensemble (129 held-out molecules). Points are R^2 on the test set with 95% bootstrap confidence intervals for the baseline (A), permutation-importance feature selection (B), and feature selection plus Gaussian-noise augmentation (C); horizontal lines mark the values reported in the original study Karimova and Senanayake (2025).

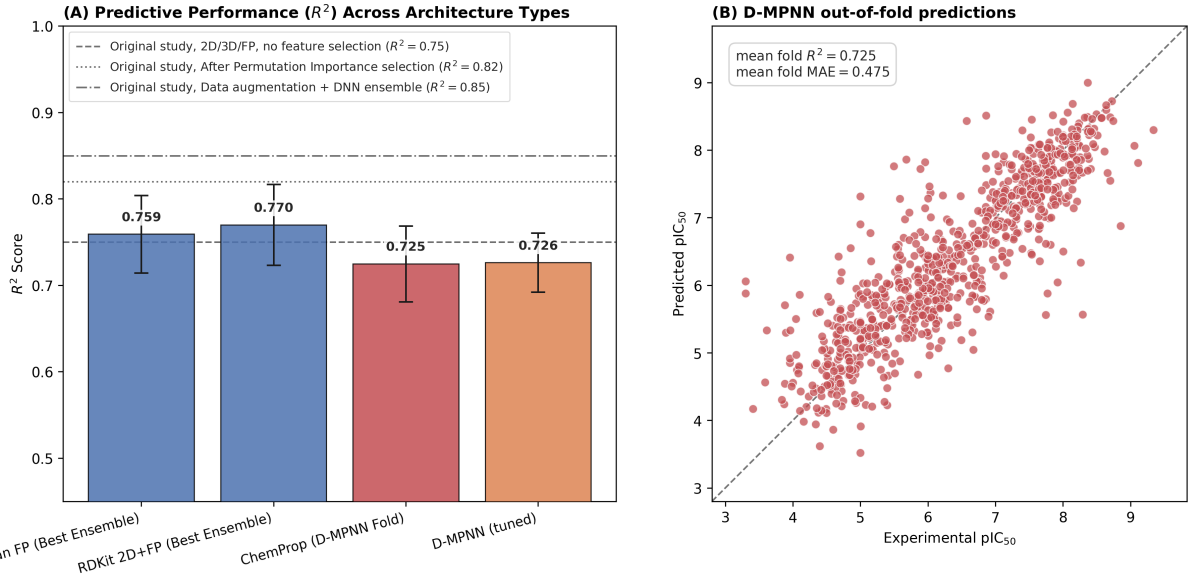


Figure 5: (A) Mean R^2 over the 25 shared outer folds for the best classical ensembles and the D-MPNN with default and tuned hyperparameters; error bars are the standard deviation across folds, and horizontal lines mark the values reported in the original study. (B) Default D-MPNN predicted versus experimental pIC_{50} for every compound, averaged over the five repeats of the cross-validation (each prediction comes from a model that did not see the compound).

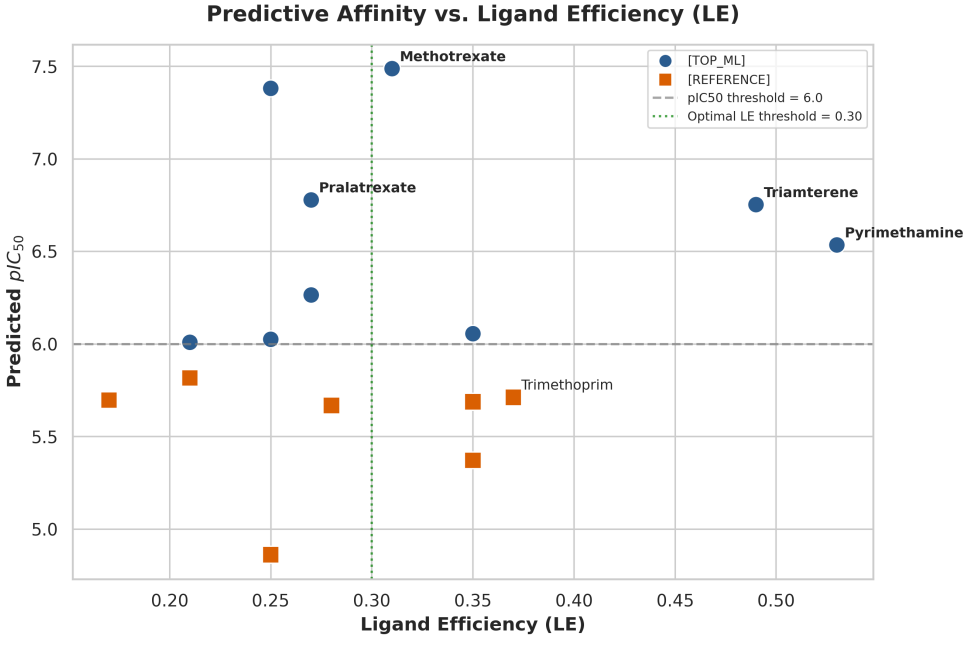


Figure 6: Predicted pIC_{50} versus ligand efficiency for the candidates advanced to docking. Blue: the ten top-ranked compounds by predicted potency; orange: reference compounds from the original study. The dashed line marks $pIC_{50} = 6.0$ and the dotted line LE = 0.30.

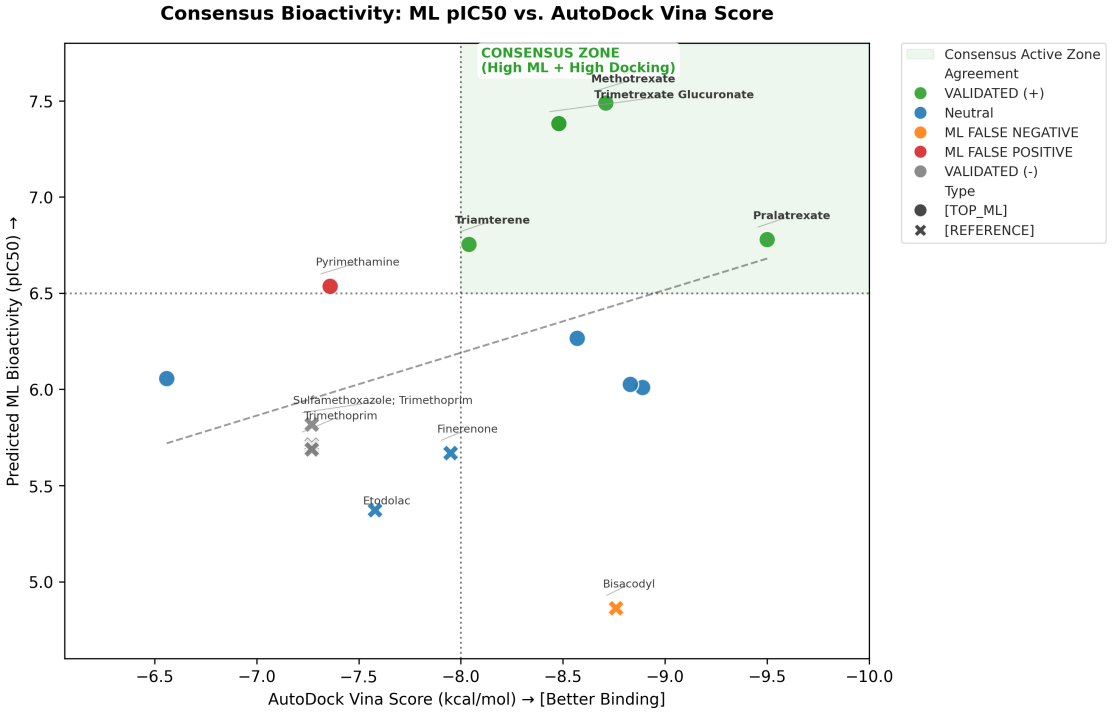


Figure 7: Predicted pIC_{50} versus AutoDock Vina score for the 17 docked candidates, coloured by agreement class. The shaded region is the consensus zone ($pIC_{50} > 6.5$ and $\Delta G < -8.0$ kcal/mol); the dashed line is a least-squares trend.

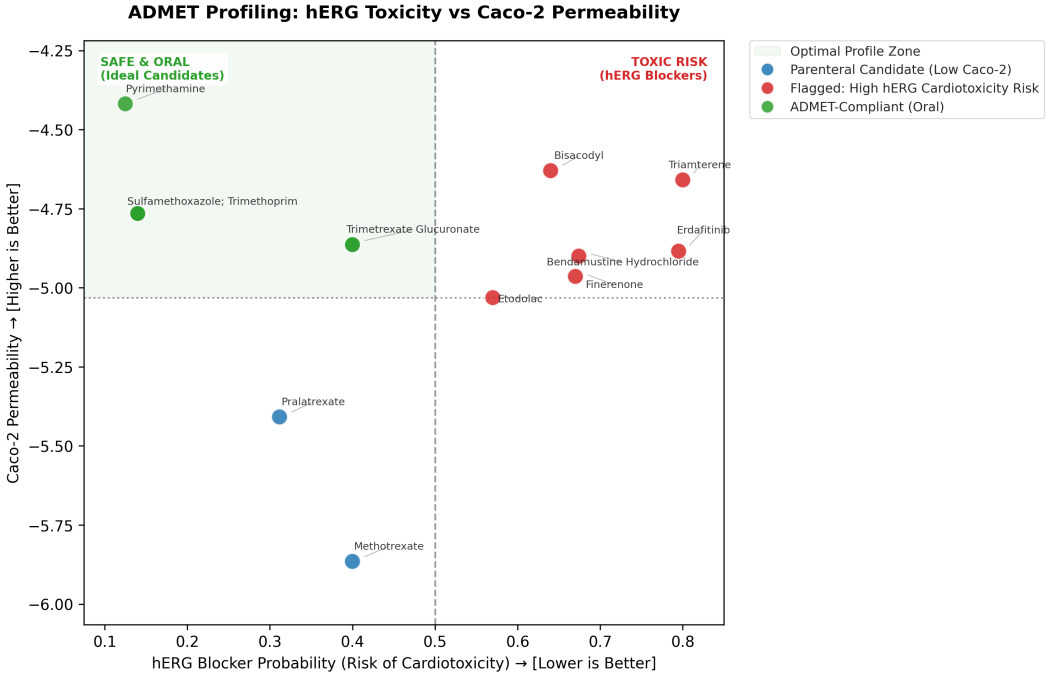


Figure 8: Predicted hERG blocking probability versus predicted Caco-2 permeability for the docked candidates that passed the Lipinski criteria. The dashed lines mark the hERG threshold and the Caco-2 threshold; the shaded quadrant combines low hERG risk with permeability classified as oral.